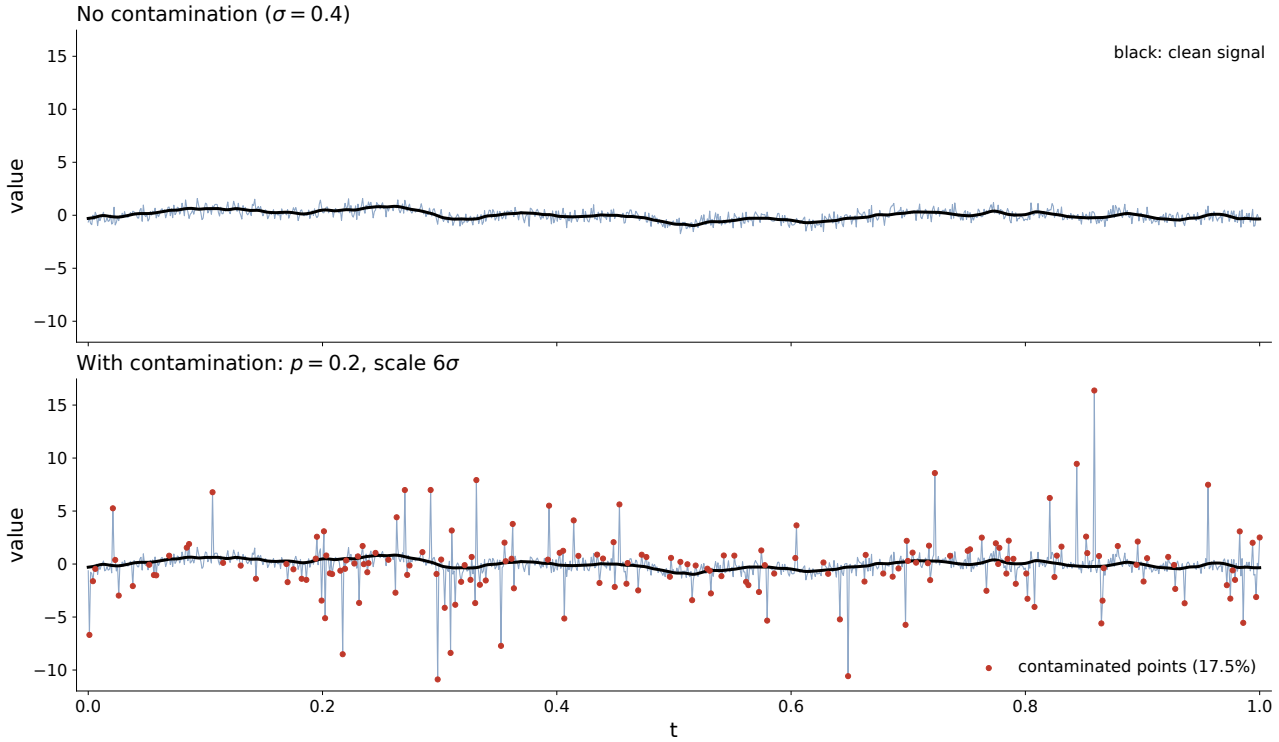
Intrinsic Robust Multiscale Filtering

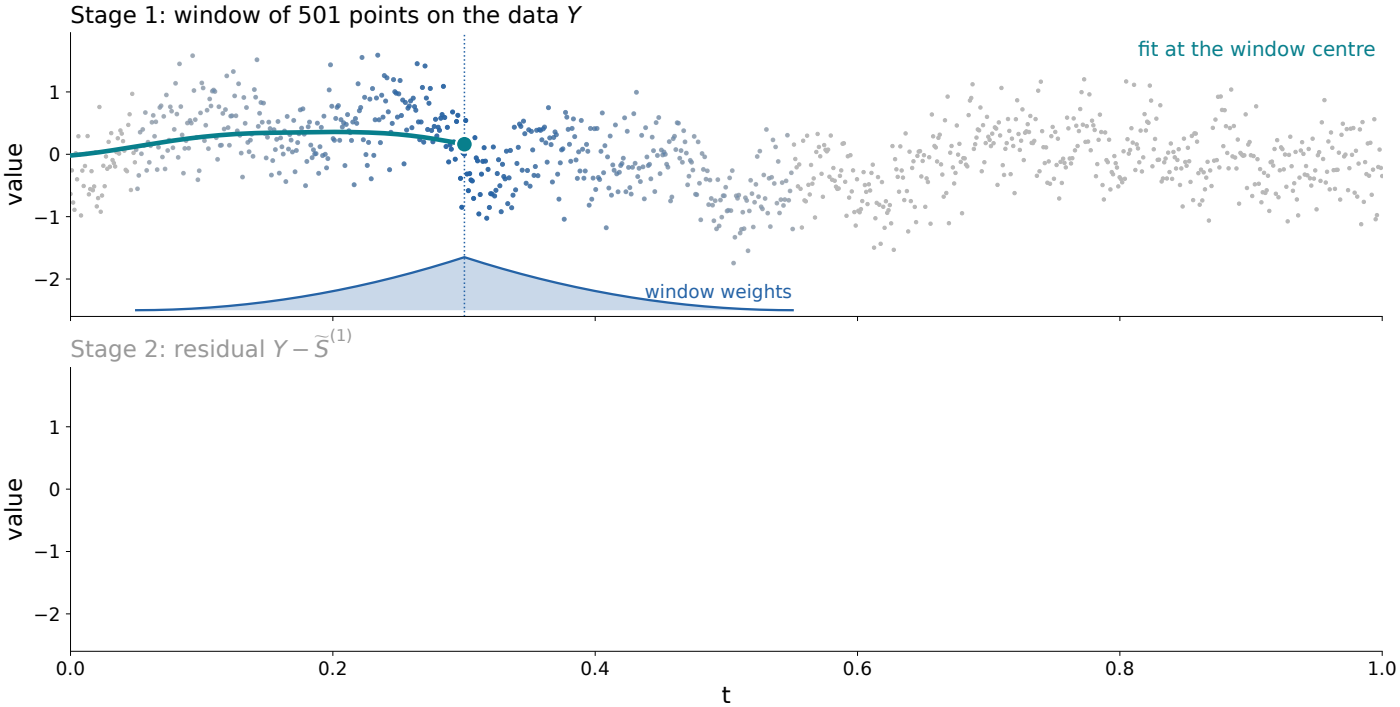
Splitting a signal into scales when the data are contaminated

The problem: signal + noise + contamination



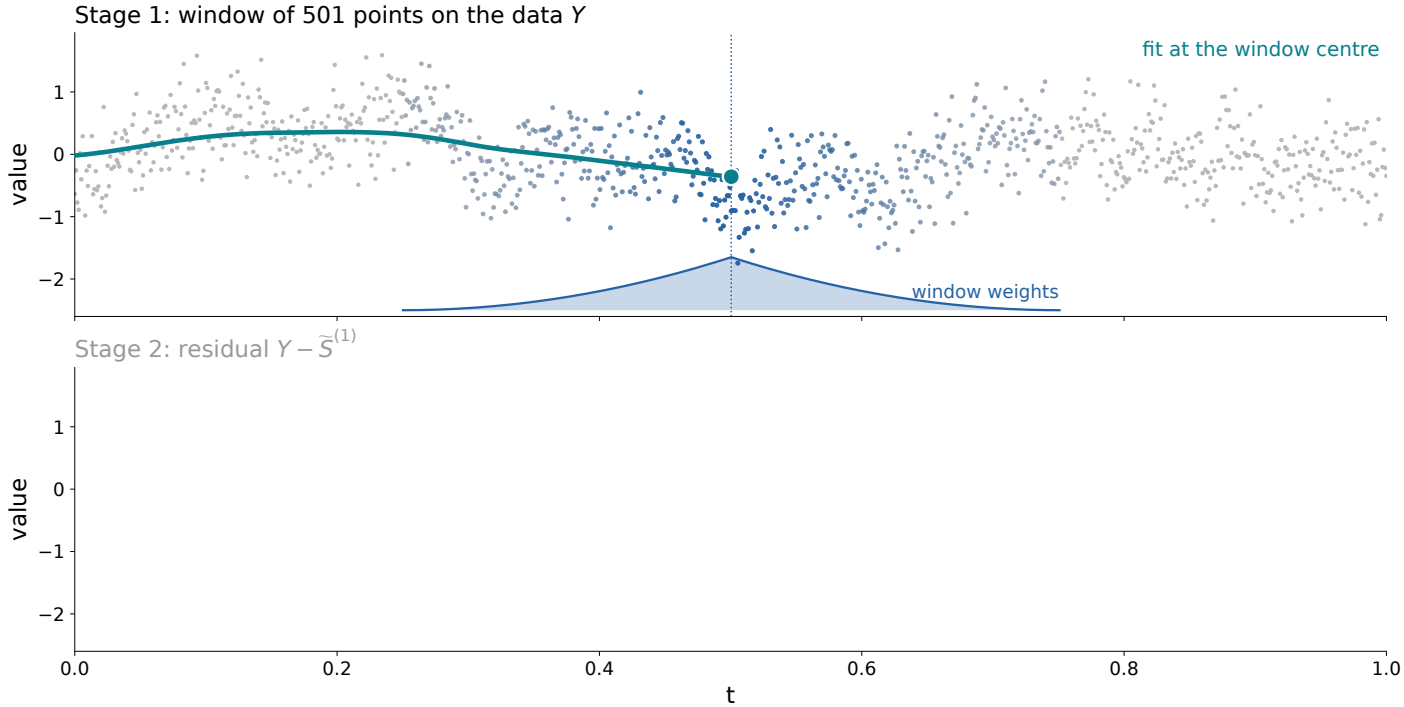
$Y_i = X_i + G_i + M_i C_i$: Gaussian noise $G_i \sim \mathcal{N}(0, \sigma^2)$; at a random fraction p of points ($M_i = 1$) a gross error C_i. Goal: recover the coarse-to-fine components of the clean signal X.

IMF in pictures: a window slides along the series



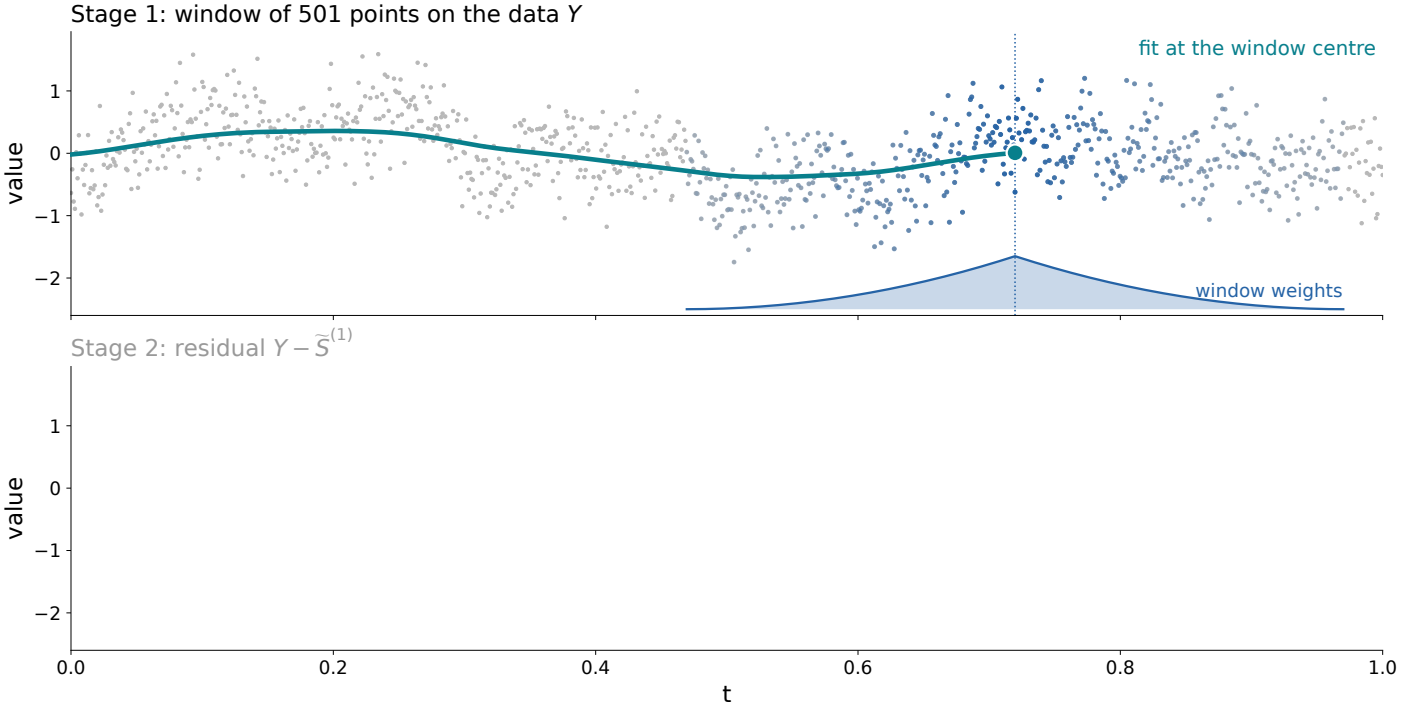
At each time t, fit one value to the points in the window around t (darker = more weight). Moving the window traces component 1.

IMF in pictures: a window slides along the series



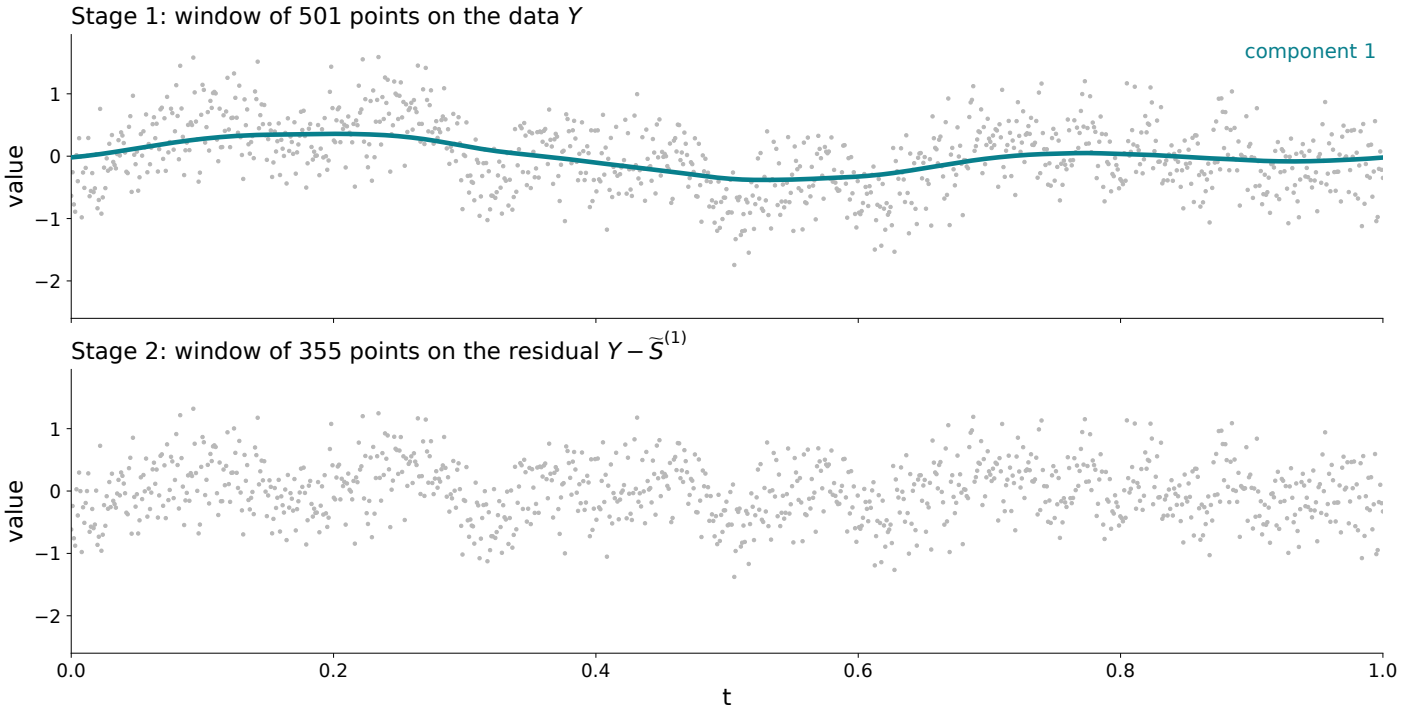
At each time t, fit one value to the points in the window around t (darker = more weight). Moving the window traces component 1.

IMF in pictures: a window slides along the series



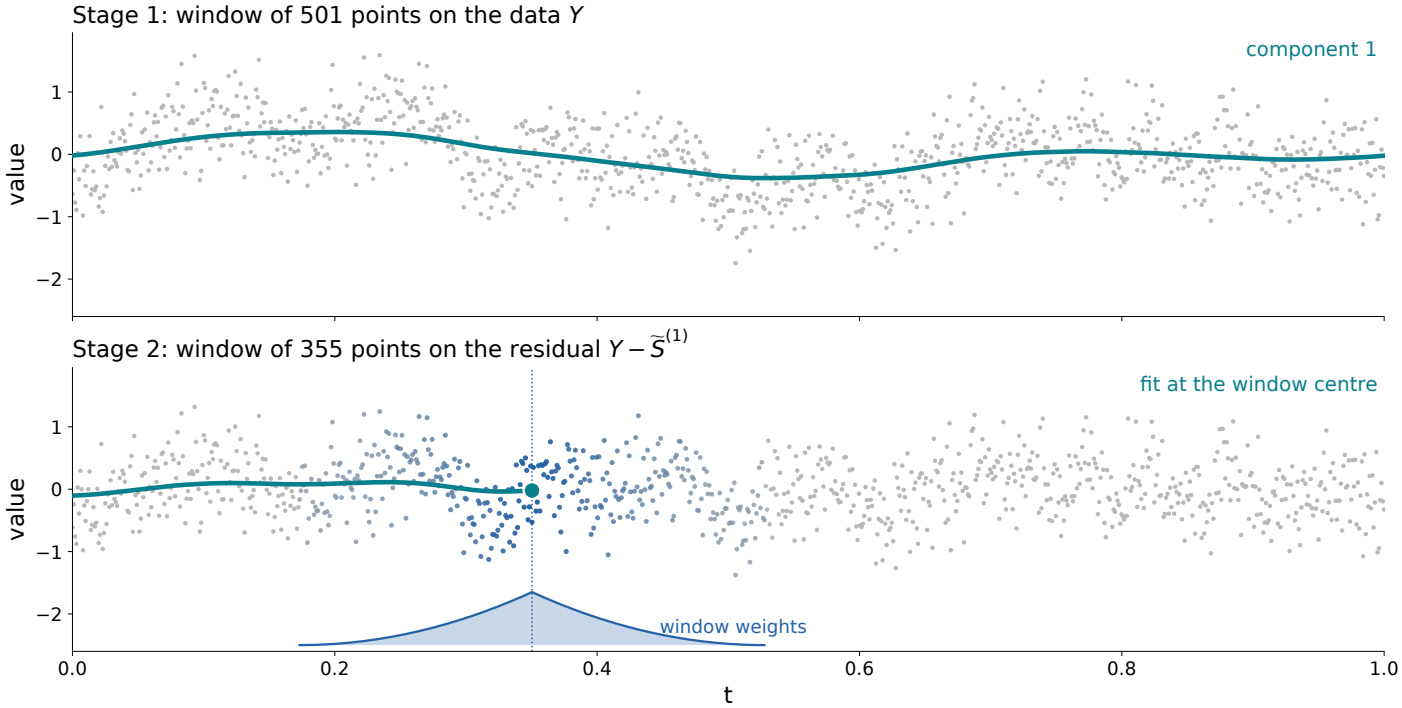
At each time t, fit one value to the points in the window around t (darker = more weight). Moving the window traces component 1.

IMF in pictures: a window slides along the series



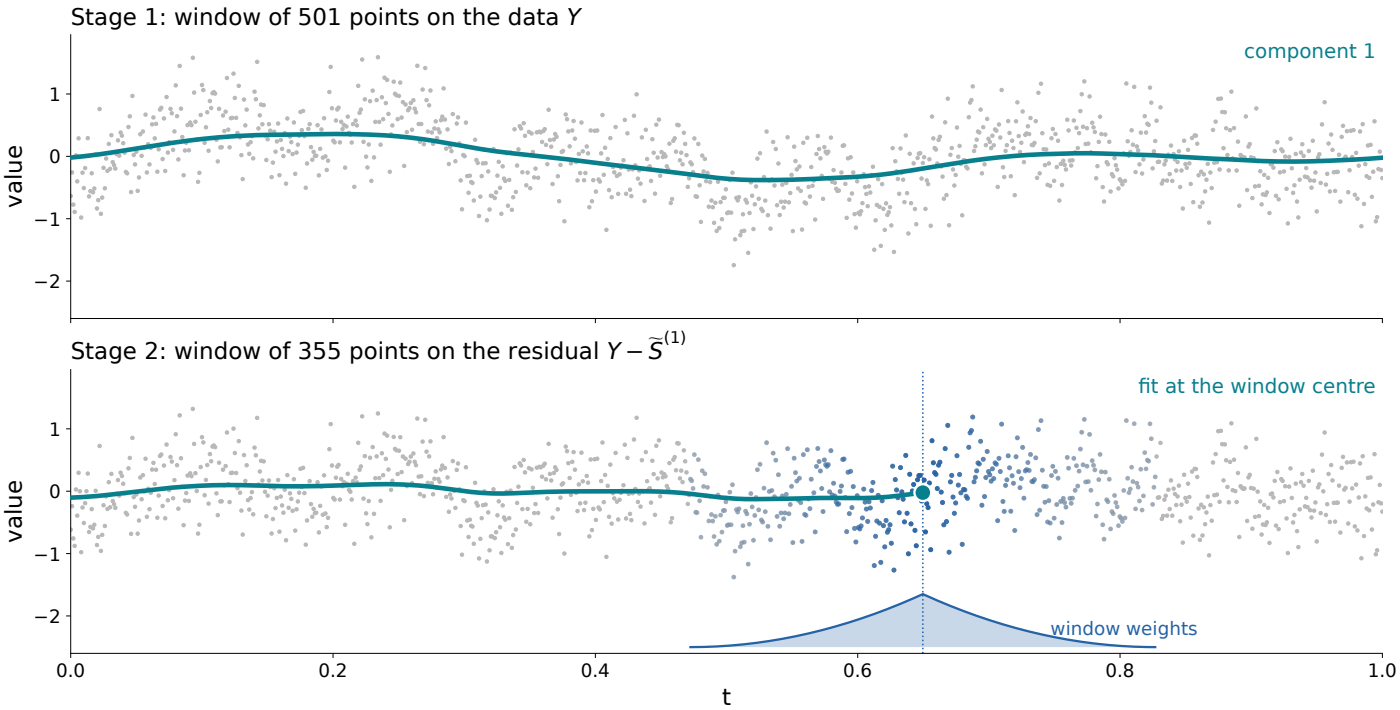
Subtract component 1. What remains is passed to the next stage.

IMF in pictures: a window slides along the series



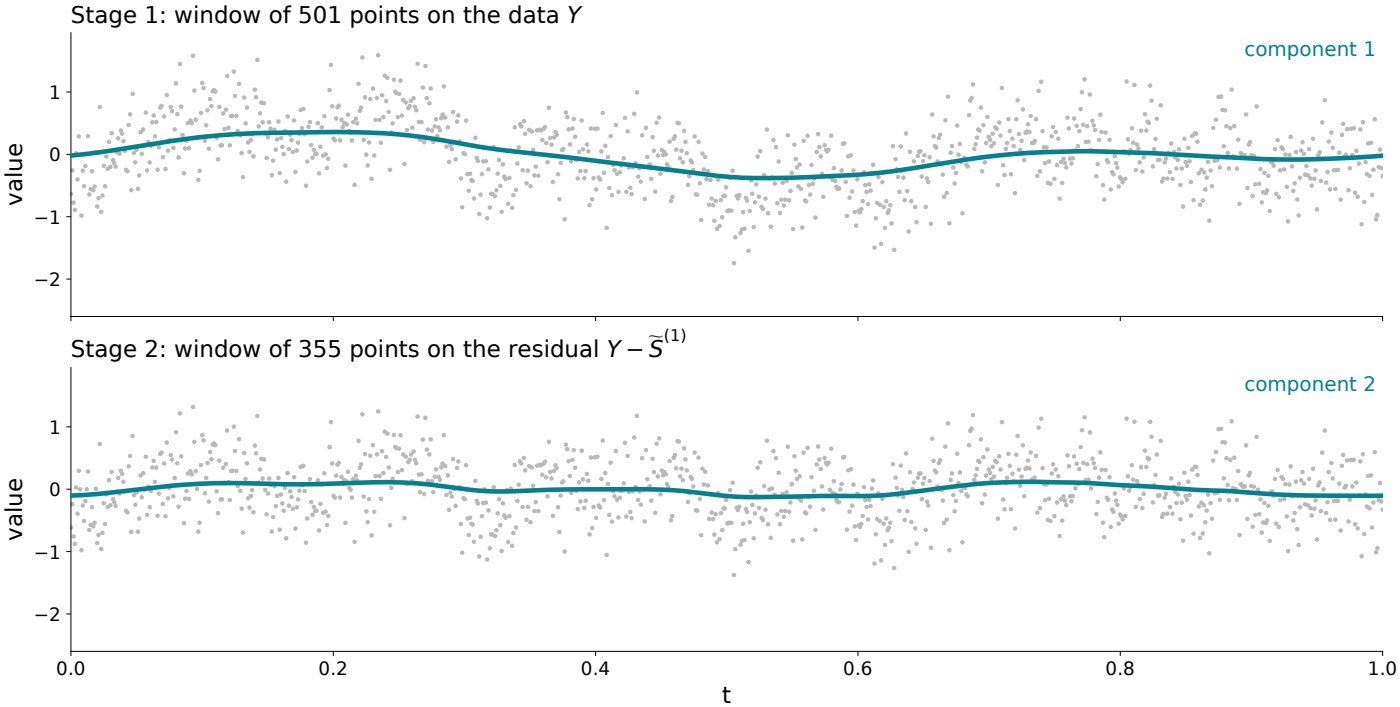
Same on the residual, with a window $a = \sqrt{2}$ times narrower: component 2.

IMF in pictures: a window slides along the series



Same on the residual, with a window $a = \sqrt{2}$ times narrower: component 2.

IMF in pictures: a window slides along the series



Repeat with ever narrower windows, $501 \to 355 \to 251 \to \cdots \to 31$ points.

IMF: peel off scales from coarse to fine

Start with $Y^{(1)} \stackrel{\text{def}}{=} Y$ and a wide window h_1. At stage k:

$$\widetilde{S}_t^{(k)} \stackrel{\text{def}}{=} \arg\min_x \sum_u \rho\big(Y_u^{(k)} - x\big)\, K\Big(\frac{t-u}{h_k}\Big), \qquad Y^{(k+1)} \stackrel{\text{def}}{=} Y^{(k)} - \widetilde{S}^{(k)}, \qquad h_{k+1} \stackrel{\text{def}}{=} \frac{h_k}{a}.$$

Linear IMF

$$\rho(r) \stackrel{\text{def}}{=} \tfrac{1}{2} r^2 \;\Rightarrow\; \text{local weighted mean}$$

Every point pulls with strength $\propto$ its distance.

Robust IMF (IRMF)

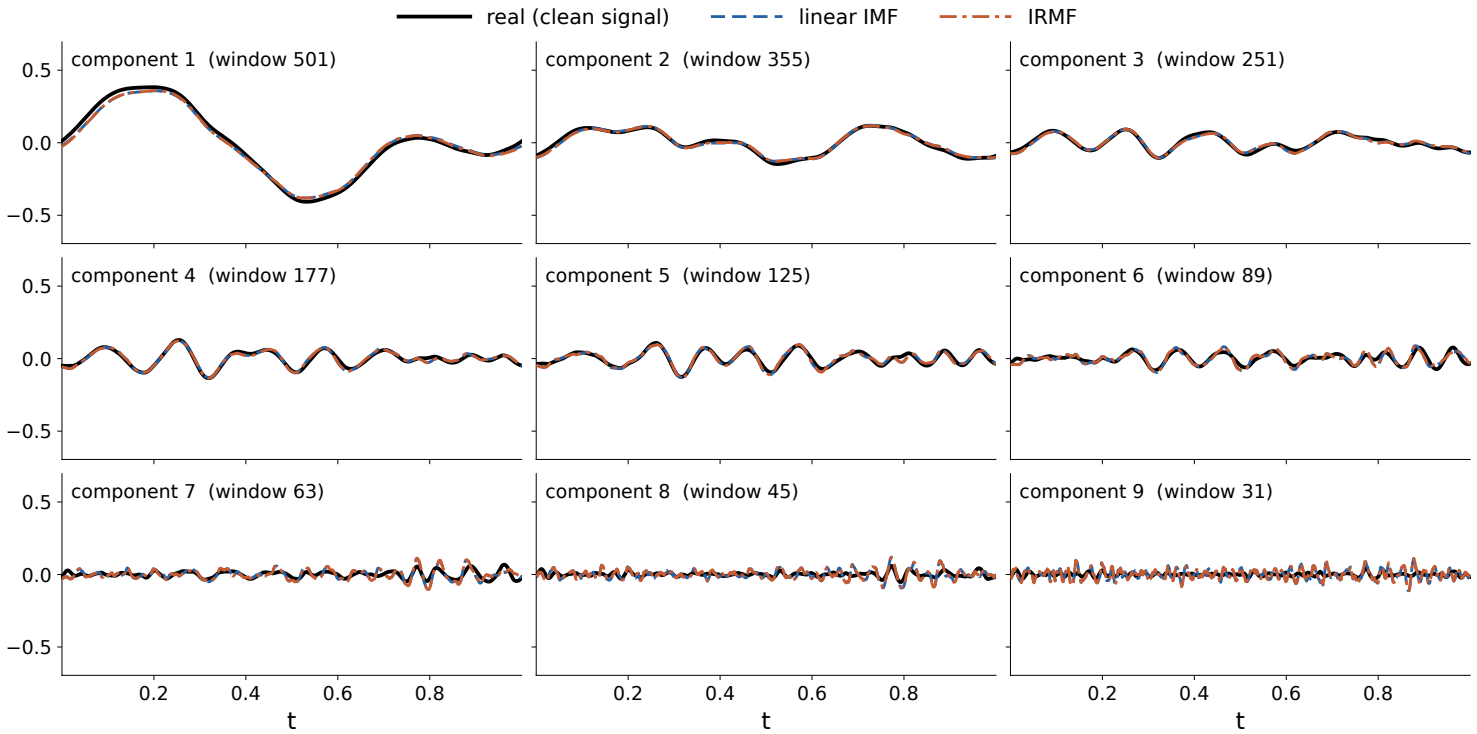
$$\rho_H(r) \stackrel{\text{def}}{=} \mathbb{E}\,|r + H\xi|, \qquad \psi_H(r) \stackrel{\text{def}}{=} \rho_H'(r), \quad |\psi_H| < 1$$

Near points pull like in the mean; far points pull with a capped strength.

Reconstruction is exact: $Y = \sum_k \widetilde{S}^{(k)} + Y^{(K+1)}$.

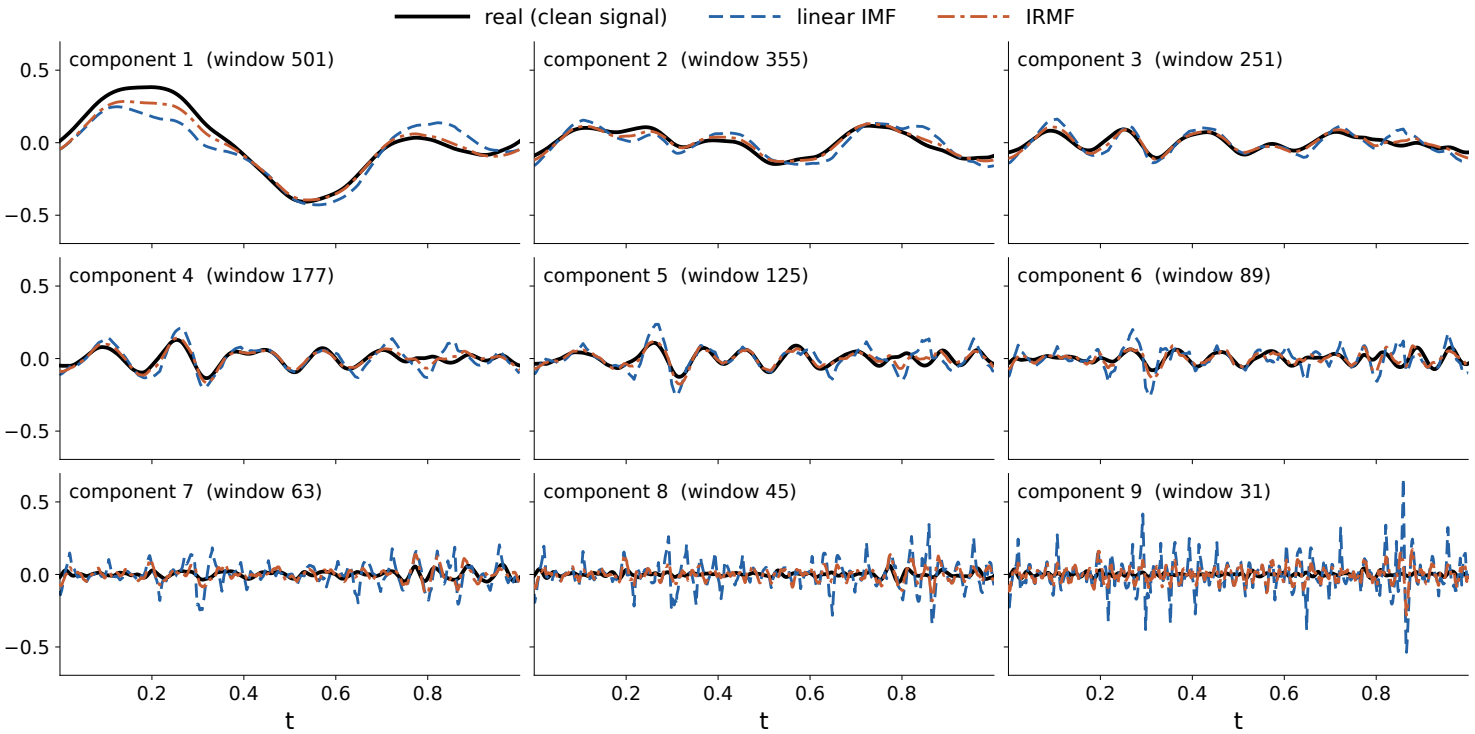
Here $a = \sqrt{2}$, nine windows from 501 to 31 points, $n = 1000$, $H = 2\sigma$; $\xi \sim \mathcal{N}(0,1)$.

No contamination: both methods recover the components



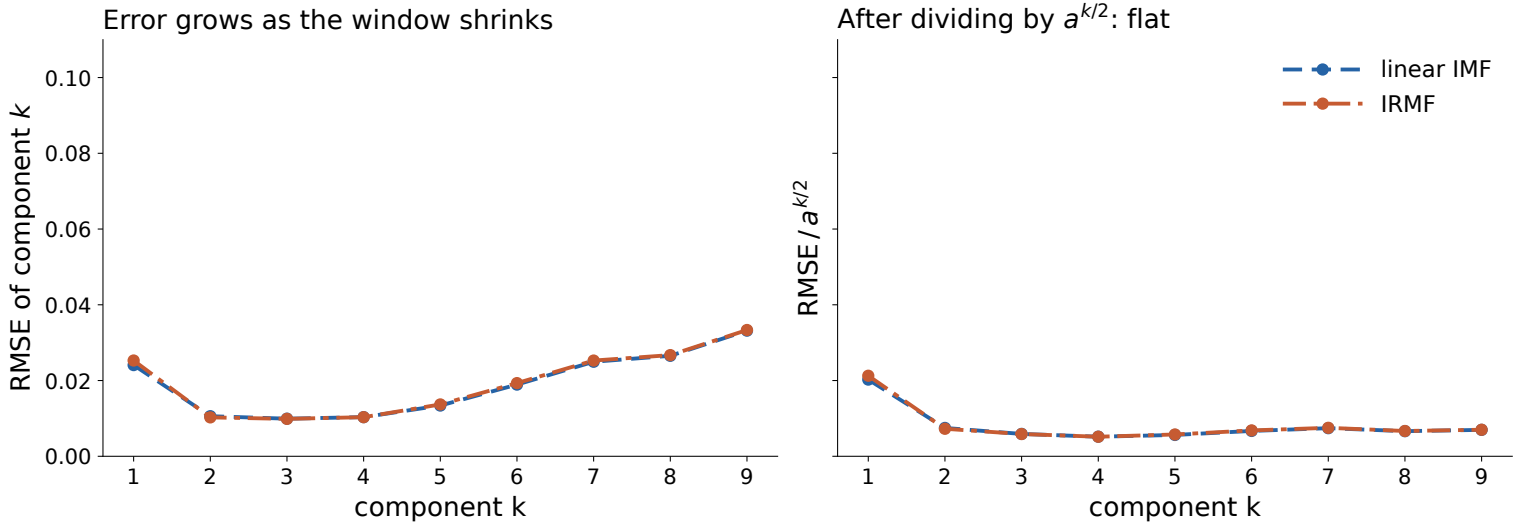
Observation in the top panel of slide 2. Real = linear IMF of the clean signal. Mean RMSE over the nine components: 0.019 for both methods.

20% contamination: IRMF stays close, linear IMF drifts



Same y-scale as the previous slide. Mean RMSE over the nine components: linear 0.068, IRMF 0.031 (−55%).

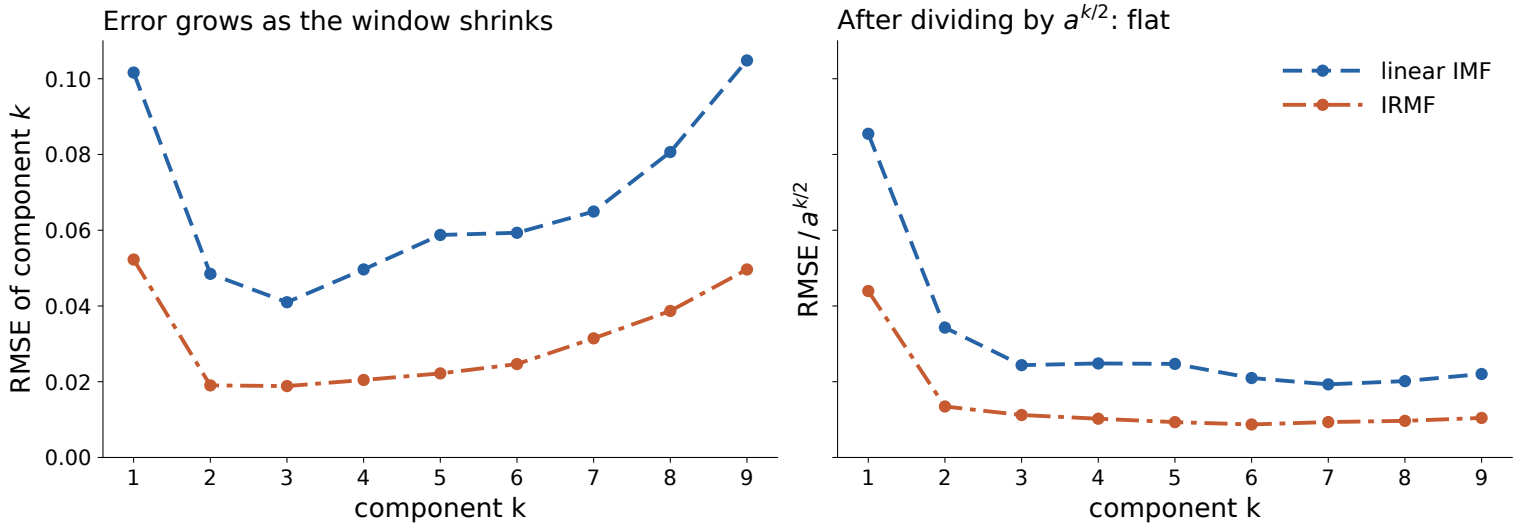
Error per component: no contamination



A window of N points averages noise down to $\sigma/\sqrt{N}$. Each stage shrinks the window by a, so the error grows by $\sqrt{a}$ per stage. Divided by $a^{k/2}$, it is flat: errors do not pile up through the recursion.

Each method compared with its own clean decomposition; same y-scale as the next slide.

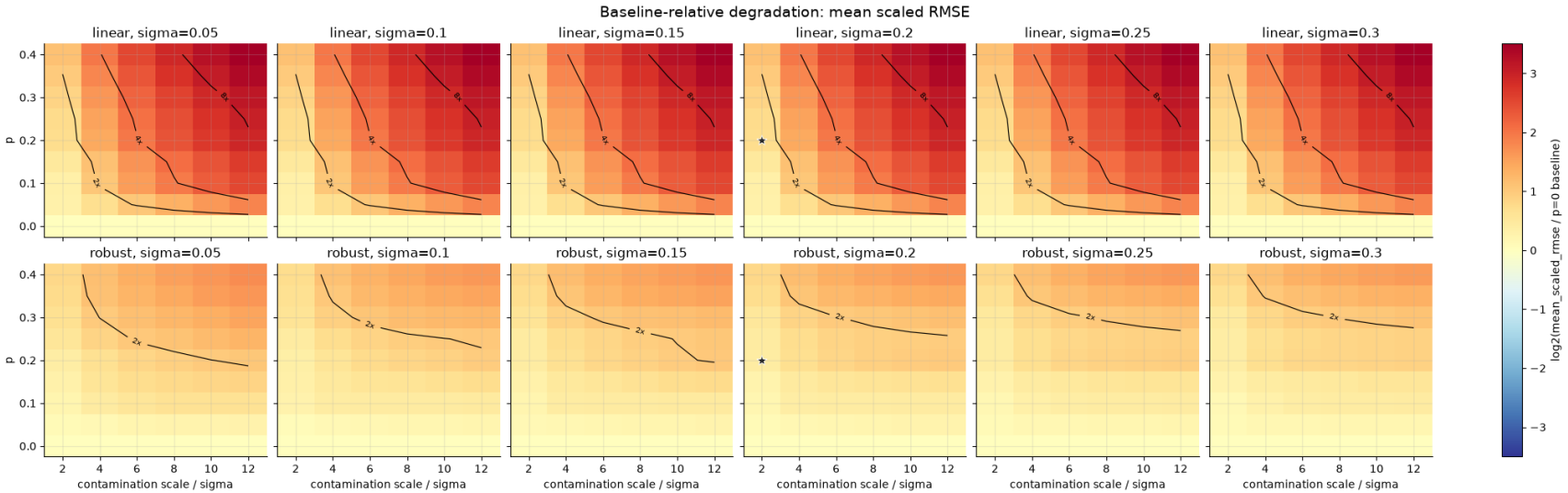
Error per component: 20% contamination



IRMF is below linear IMF at every component, and after scaling its error is again flat.

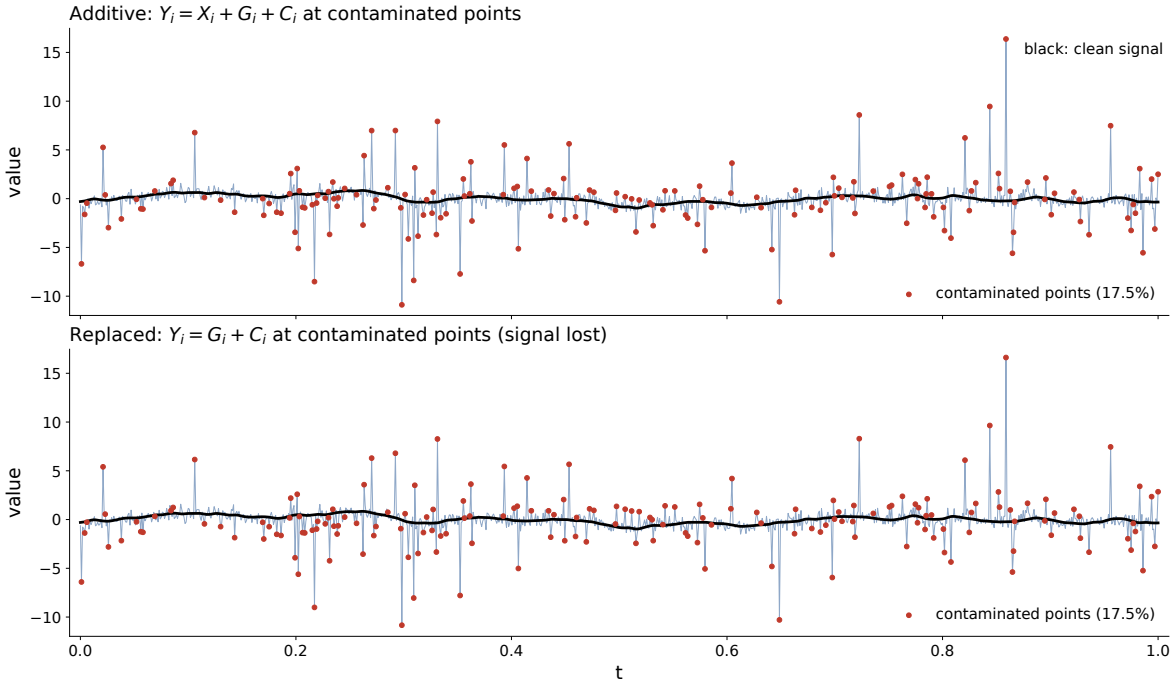
Mean RMSE over components: linear 0.068, IRMF 0.031.

Sweep: linear IMF degrades up to 8×, IRMF about 2×



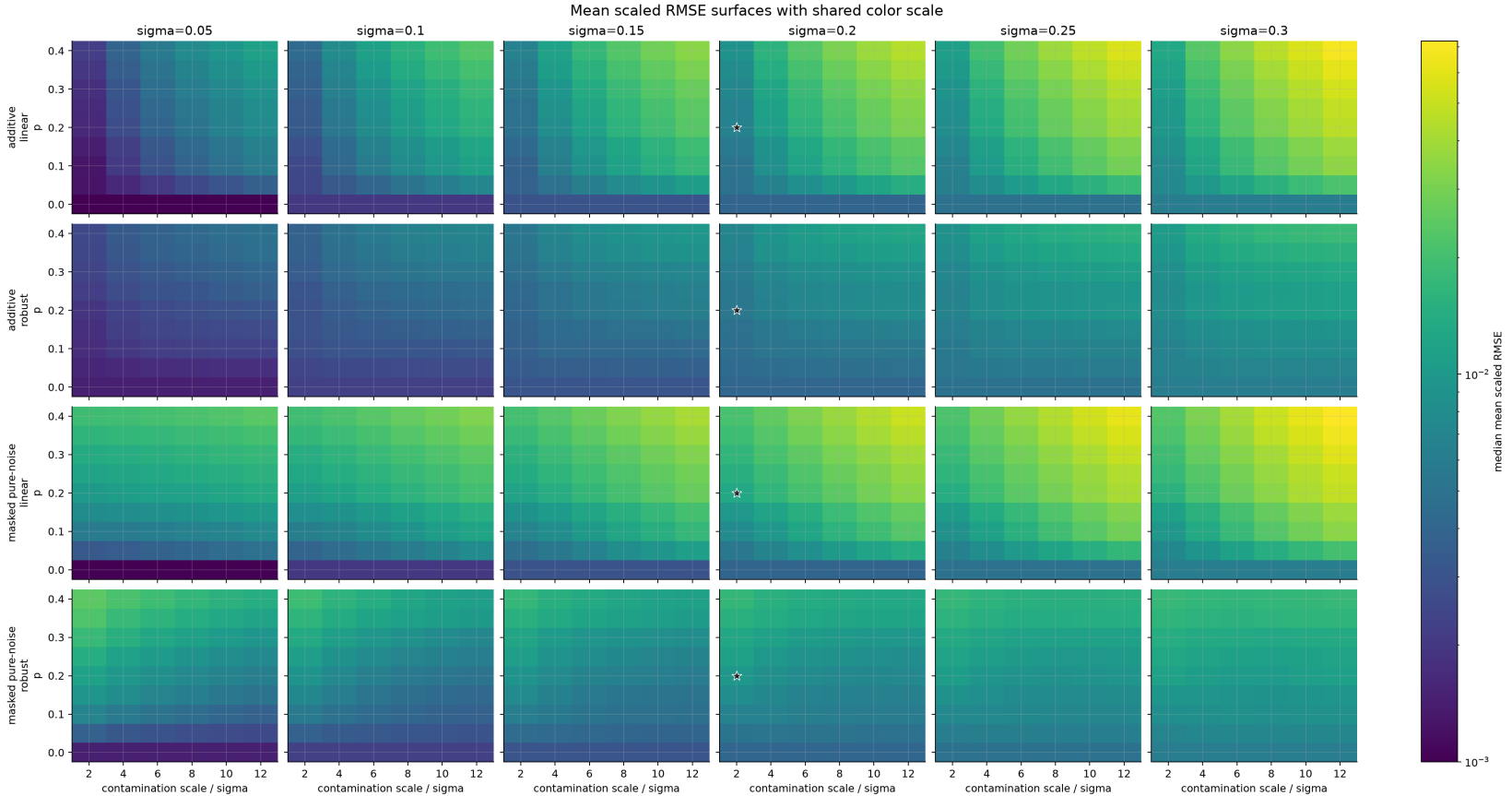
Additive contamination. Each method's error divided by its own error without contamination; 324 settings × 3 repeats. Contours mark 2×, 4×, 8×.

A second model: contamination replaces the signal



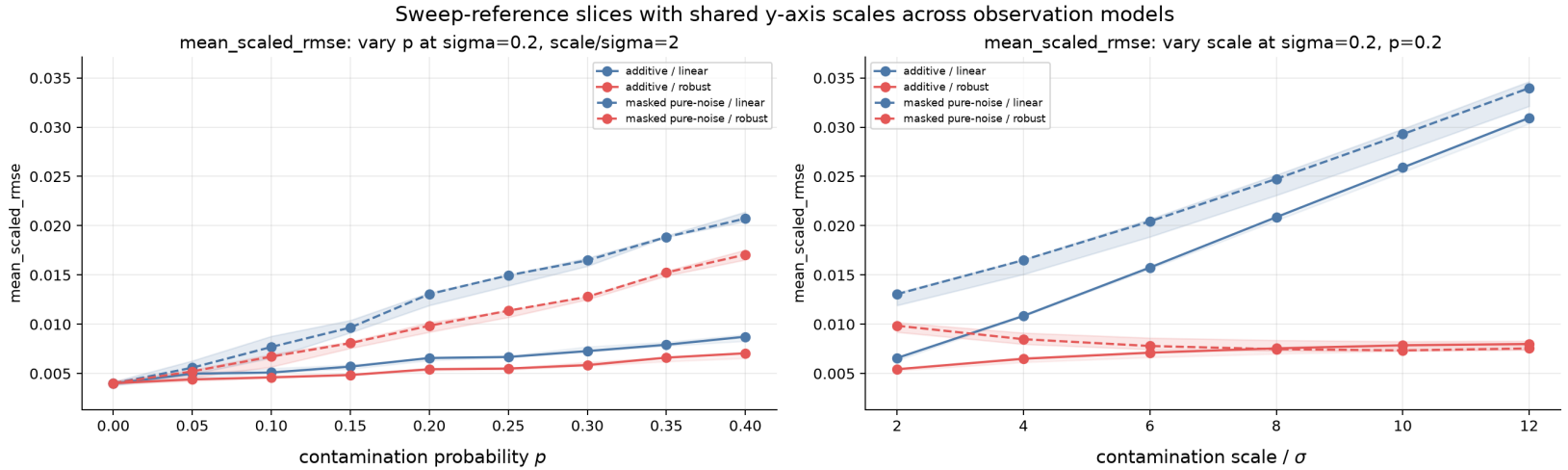
Same draws in both panels. In the replaced model the contaminated points carry no information about X.

Sweep: IRMF error barely depends on contamination size



Rows: additive / replaced (masked) contamination × linear / IRMF. One colour scale; 324 settings × 3 repeats.

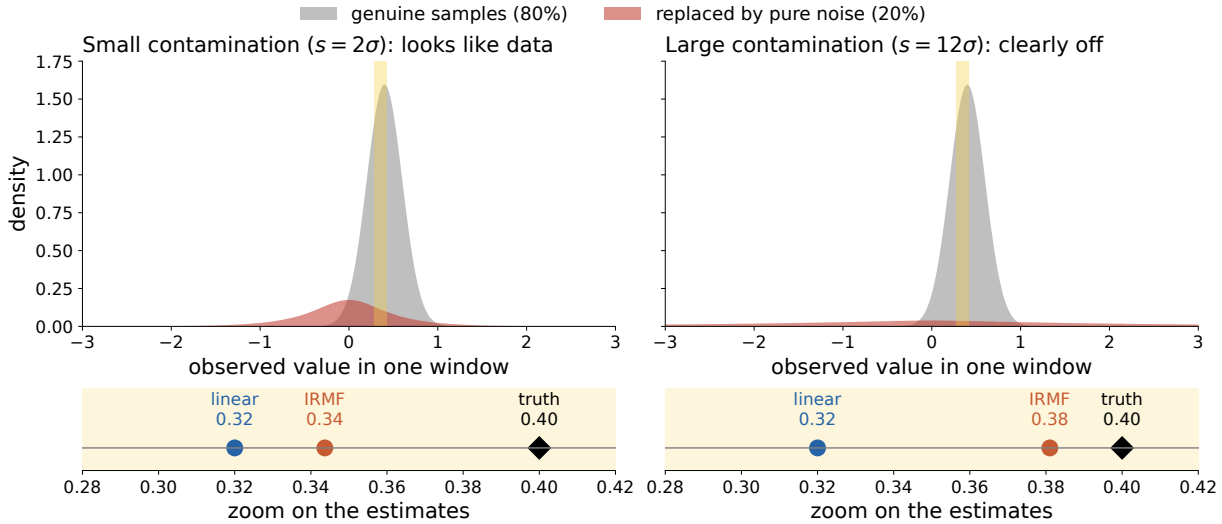
Contamination probability and size at $\sigma = 0.2$



Right: when contamination *replaces* the signal (dashed red), IRMF's error decreases as the contamination gets larger.

Median and IQR over 3 repeats. Confirmed with 100 repeats (backup): 0.0093 at $2\sigma \to 0.0082$ from 6σ on.

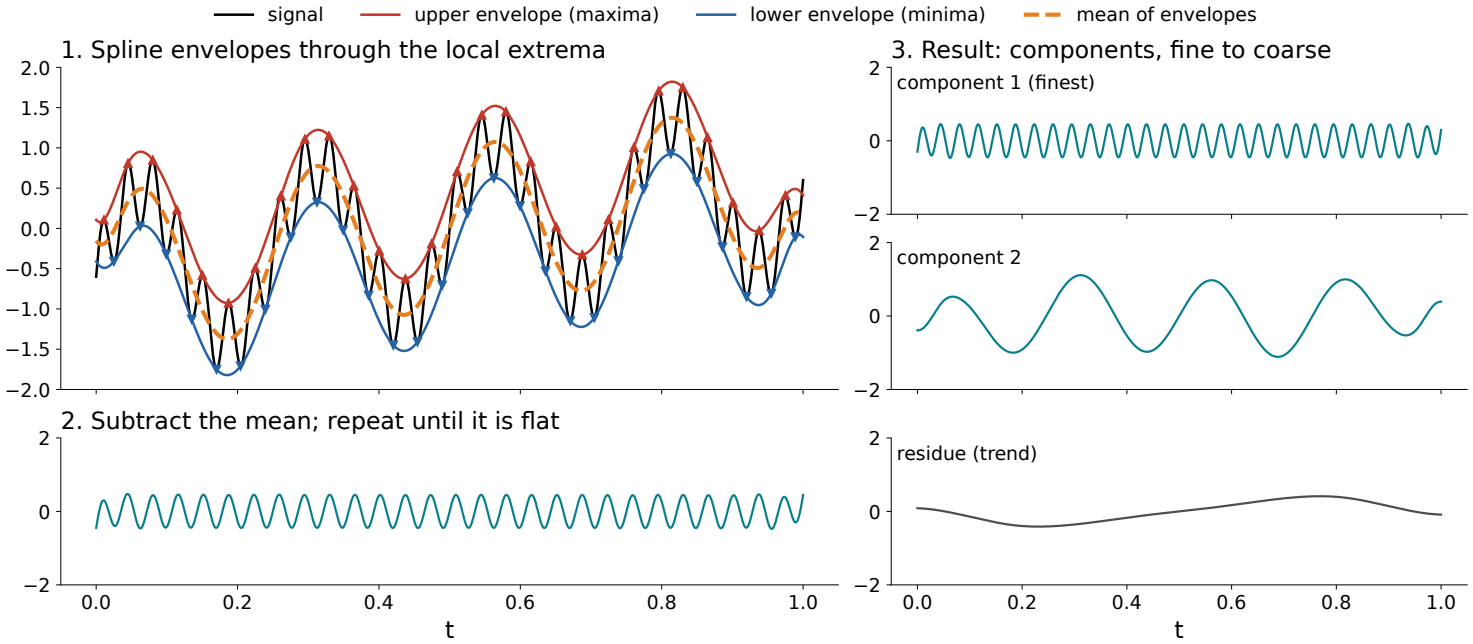
Why larger contamination can help IRMF



A replaced point carries no signal. Linear IMF always averages it in: 0.8×truth. IRMF caps each point's pull. Small contamination looks like data and drags it toward 0; large contamination sits far on both sides, the capped pulls cancel, and IRMF effectively ignores those points.

One window: true value 0.4, $\sigma = 0.2$, $H = 0.4$, 20% replaced by $G + C$, $C = \pm s \cdot \mathrm{Exp}(1)$; population estimates.

EMD: the standard data-driven decomposition



Empirical mode decomposition (Huang et al., 1998) repeats steps 1–2 (*sifting*) to peel off the finest component, then starts again on the rest.

No statistical model behind the steps; noise and spikes move the extrema, so scales can mix across components.

Real signals: two audio datasets

BirdCLEF 2023

16 bird species
12,000 frames from 1,280 recordings
451 recordists

ESC-50 animals

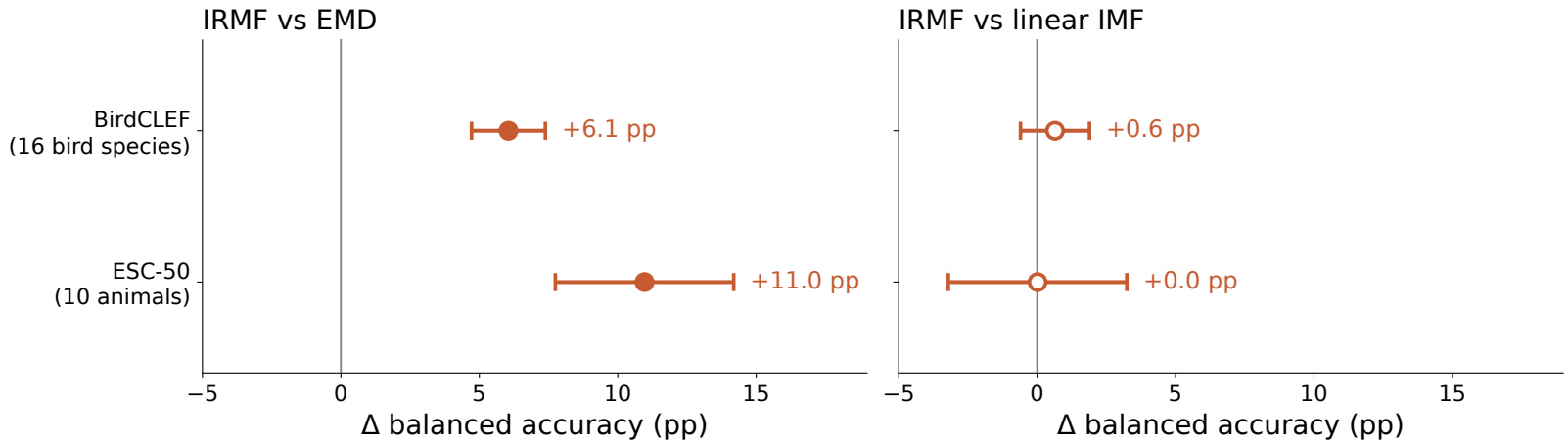
10 animal classes (dog, rooster, frog, …)
5,100 frames from 400 clips
157 uploaders

Each frame: 2,048 samples (93 ms at 22 kHz).

Pipeline: frame $\rightarrow$ EMD / linear IMF / IRMF $\rightarrow$ simple statistics of each component (energy, spread, zero crossings, spectral centroid, …) $\rightarrow$ linear classifier.

Cross-validation folds hold out whole recordists / uploaders, so the same recording never appears in training and test.

Real signals: IMF components beat EMD as features



IRMF beats EMD by +6.1 pp (BirdCLEF) and +11.0 pp (ESC-50) balanced accuracy. Against linear IMF there is no significant difference, so the gain over EMD comes from the IMF construction.

Fixed classifier: logistic regression (BirdCLEF), linear SVM (ESC-50); 5 grouped folds × 3 seeds; Nadeau–Bengio corrected 95% CIs; filled = Holm $p < 0.05$.

Software: `pimf`

NumPy-only Python library; linear and robust fits, parallel windows.

```
pip install pimf

from pimf import IMF, SmoothAbs
res = IMF(SmoothAbs(H=0.4), solver="newton").decompose(y, k_max=9)
res.imfs, res.residual, res.stages
```

Why it works (1): the fit and its population target

For one component, with kernel weights $w_{u,t} \stackrel{\text{def}}{=} K_h(t-u)$:

$$\widetilde{S}_t \stackrel{\text{def}}{=} \arg\inf_z \sum_u \rho(Y_u - z)\, w_{u,t}, \qquad S_t \stackrel{\text{def}}{=} \arg\inf_z \sum_u \mathbb{E}\,\rho(Y_u - z)\, w_{u,t}.$$

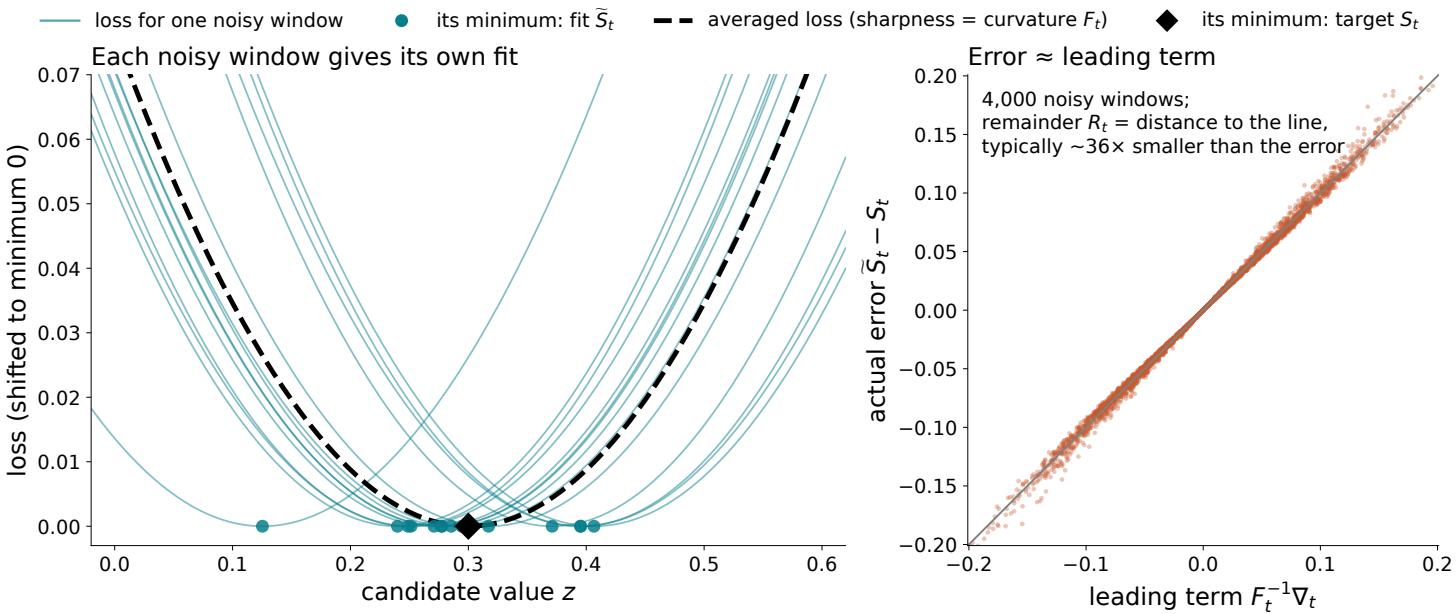
$\widetilde{S}_t$ is the fit to the data; S_t is its *population target*, the fit after averaging out the noise. With $\phi(z) \stackrel{\text{def}}{=} \mathbb{E}\,\rho(z+\varepsilon)$, curvature and centred score at the target are

$$F_t \stackrel{\text{def}}{=} \sum_u \phi''(X_u - S_t)\, w_{u,t}, \quad D_t \stackrel{\text{def}}{=} F_t^{1/2}, \qquad \nabla_t \stackrel{\text{def}}{=} \sum_u \big\{\rho'(Y_u - S_t) - \mathbb{E}\,\rho'(Y_u - S_t)\big\} w_{u,t}.$$

$$\widetilde{S}_t - S_t = \underbrace{F_t^{-1}\nabla_t}_{\text{leading random term}} + \underbrace{R_t}_{\text{remainder}}$$

The identity defines R_t. Independent errors with $\mathbb{E}|\varepsilon_u| < \infty$; ρ convex, C^3, with $|\rho'| \le L_1$, $0 \le \rho'' \le L_2$, $|\rho'''| \le L_3$ (for ρ_H: $L_1 = 1$).

Why it works (1) in pictures: fit vs. target



Left: each noisy window has its own minimum $\widetilde{S}_t$; they scatter around the minimum S_t of the averaged loss. Right: the error is almost exactly the leading term $F_t^{-1}\nabla_t$; the remainder R_t is tiny.

One IRMF window of 15 points, $\sigma = 0.2$, $H = 0.4$; 14 draws on the left, 4,000 on the right (median $|R_t| = 0.0013$ vs. median error 0.046).

Why it works (2): Proposition 4.1

Proposition 4.1. Suppose that the conditions above hold and that $\delta < 1$ and $\dfrac{C_1 C_3}{D_{\min}(1-\delta)^2} < \dfrac{4}{9}$. Then there is a random set $\Omega(x)$ with $\mathbb{P}\{\Omega(x)\} \geq 1 - e^{-x}$ such that

$$\sup_{t \in \mathcal{T}_n} \left| D_t \left(\widetilde{S}_t - S_t - F_t^{-1} \nabla_t \right) \right| \leq \frac{1}{D_{\min}} \left\{ \frac{9 C_1^2 C_3}{8(1-\delta)^3} + \frac{C_1 C_2}{1-\delta} \right\}.$$

On the same event, every empirical minimizer is unique and $\sup_{t \in \mathcal{T}_n} D_t |\widetilde{S}_t - S_t| \leq \dfrac{3 C_1}{2(1-\delta)}$.

where, with $z_x \stackrel{\text{def}}{=} x + \log(4m)$, $Q_t^2 \stackrel{\text{def}}{=} \sum_u w_{u,t}^2$, $k_t \stackrel{\text{def}}{=} \max_u w_{u,t}$, $W_t \stackrel{\text{def}}{=} \sum_u w_{u,t}$:

$$C_1 \stackrel{\text{def}}{=} \max_t \frac{L_1}{D_t} \left\{ Q_t \sqrt{2 z_x} + \tfrac{2}{3} k_t z_x \right\}, \quad C_2 \stackrel{\text{def}}{=} \max_t \frac{L_2}{D_t} \left\{ Q_t \sqrt{2 z_x} + \tfrac{1}{3} k_t z_x \right\}, \quad C_3 \stackrel{\text{def}}{=} \max_t \frac{L_3 W_t}{D_t^2}, \quad \delta \stackrel{\text{def}}{=} \frac{C_2}{D_{\min}}.$$

Lemma 4.2: on $\Omega(x)$, C_1 bounds the centred score, C_2 the curvature fluctuation and C_3 the third derivative, simultaneously for all $t \in \mathcal{T}_n$ (m evaluation points). Conditions: those on the previous slide.

Why it works (3): later components

- **Linear IMF.** Component k sees the noise through $e_k = W_{h_k} \prod_{j<k} (I - W_{h_j})\, \varepsilon$. For a positive-definite kernel each factor $I - W_h$ is a contraction, so the error of component k stays at the noise level of its own window, $\sigma/\sqrt{n h_k}$ — the $a^{k/2}$ growth on slides 7–8.
- **IRMF: even the second component is work in progress.** Its input residual $Y - \widetilde{S}^{(1)}$ has noise that is no longer independent or Gaussian, so the first-component argument does not apply directly.
- **Draft idea.** Replace Bernstein by Gaussian concentration for Lipschitz functions; the hope is that the one-step rate $\sqrt{z_x/(n h_2)}$ survives, up to a constant.
- **Open.** Components $k \geq 2$ for IRMF, and theory under contamination (current results assume Gaussian noise).

With the periodic boundary all W_h commute, so $\|e_k\| \leq \|W_{h_k} \varepsilon\|$. The kernel matters: $\|I - W_h\| \leq 1$ holds for $(1 - |u|)_+^2$ (used here), not for Epanechnikov $\frac{3}{4}(1 - u^2)_+$.

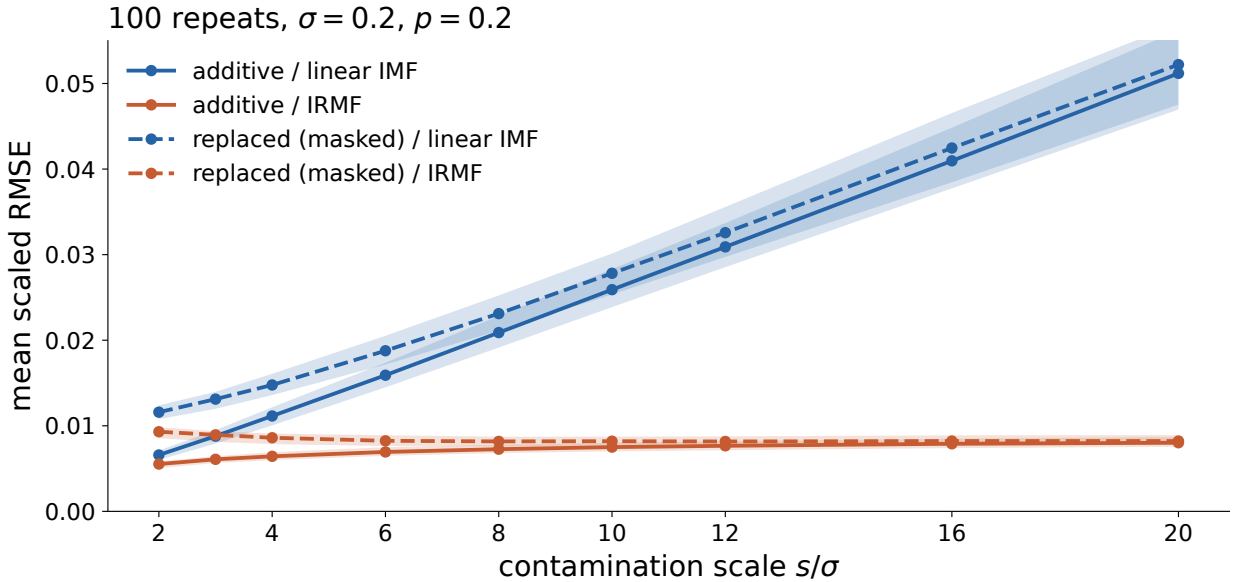
Summary

- IMF splits a signal into scales by repeated local fits on residuals; IRMF keeps the recursion but caps how hard any single point can pull.
- Without contamination the two methods agree. With contamination IRMF follows the real components, and its error stays flat as the contamination grows: -75% vs. linear IMF at 12σ.
- When contamination replaces the signal, larger contamination even helps IRMF: clearly-off points are effectively ignored, subtle ones are not.
- On real audio (BirdCLEF, ESC-50), IMF components beat EMD features by 6–11 pp.
- For the first component the fit concentrates around its population target (Proposition 4.1); later components are work in progress.

-75%: mean scaled RMSE, 100 repeats, $\sigma = 0.2$, $p = 0.2$, both contamination models.

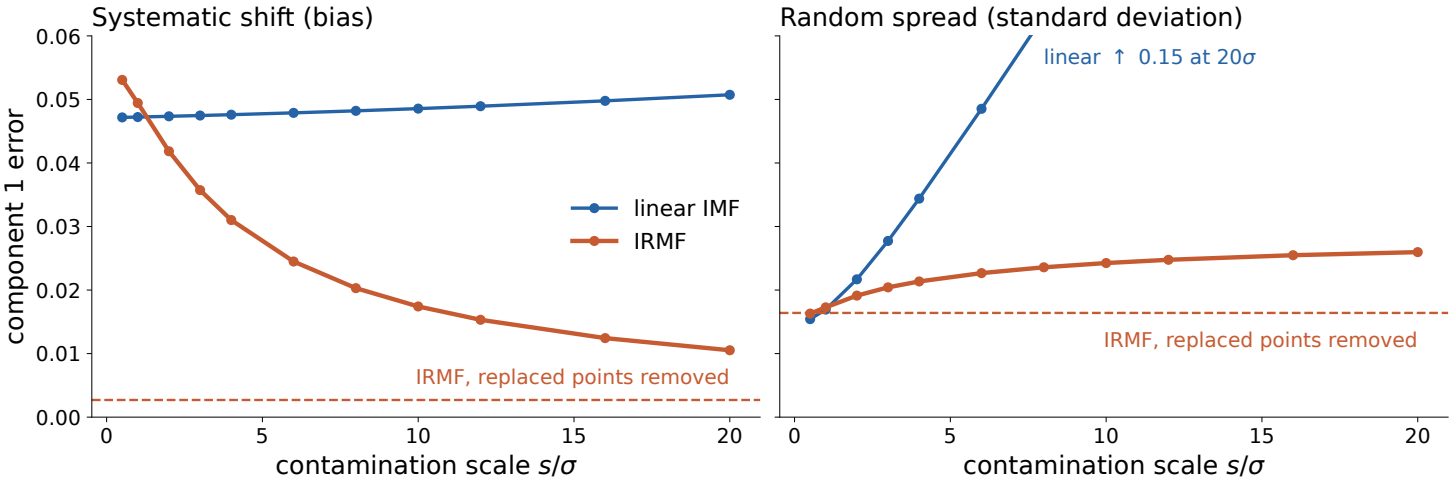
Backup

Backup: the sweep slice with 100 repeats



Same metric as slide 12 (mean over components of RMSE/$a^{k/2}$); median and IQR over 100 independent repeats.

Backup: component 1, replaced contamination



Error split over 100 repeats into a *systematic shift* (the average error) and a *random spread* around it. IRMF's shift shrinks toward the value obtained when the replaced points are removed; its spread levels off. Linear IMF keeps a fixed shift (20% of the signal is missing) and its spread grows with contamination size.

Backup: implementation details

- Windows $501, 355, 251, 177, 125, 89, 63, 45, 31$; periodic boundary; same weights for both methods.
- Implemented weights $w(u) \propto (1 - |u|)_+^2$ (the review calls them Epanechnikov).
- IRMF: $H = 2\sigma$, lookup-table score, gradient descent, at most 60 steps, tolerance 10^{-6}.
- Contamination $C_i = \pm s \cdot \mathrm{Exp}(1)$ with a random sign; replaced (masked) model: $Y_i = G_i + C_i$ at contaminated points.
- Slides 2–3, 5–8, 10: review signal and seed 777, $\sigma = 0.4$, $p = 0.2$, contamination scale 6σ.
- 100-repeat Monte Carlo: $\sigma = 0.2$, $p = 0.2$, $H = 0.4$, $s/\sigma \in \{0.5, \ldots, 20\}$, notebook code unchanged.